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**3rd Sem / Automobile, Mechanical Engg, Mechanical
(Tool & Die Design), Agri, Chem, P&P, Civil, Plastic,
Prod, CNC, CAD/CAM, Found. & Forg, Metallurgy ,
Print Making Tech., Mech (Ad Manu.Tech.) Mech.
Engg(Fabrication Tech). Mech. Engg .(Prod.)**

**Subject : Basics of Electrical and Electronics
Engineering**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Unit of Voltage is (CO1)

- a) Ohm b) Ampere
- c) Volt d) Second

Q.2 Frequency of a.c. supply in INDIA is (CO1)

- a) 50 Hz b) 60Hz
- c) 0 Hz d) 20 Hz

Q.3 MCB is used for protection against (CO6)

- a) Leakage current b) Earth fault
- c) High current d) Lightning

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- Q.4 Need of starter to start motor is to (CO5)
- a) Provide starting torque
 - b) Limit circulating current
 - c) Limit starting current
 - d) None
- Q.5 Diode is used to (CO7)
- a) Convert a.c. into d.c.
 - b) Convert d.c. into a.c.
 - c) Amplify given signal
 - d) Control electric drive
- Q.6 In star connection (CO1)
- a) $V_L = V_{Ph}$
 - b) $V_L = \sqrt{3} V_{Ph}$
 - c) $V_L = V_{Ph} / \sqrt{3}$
 - d) $3V_L = V_{Ph}$

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Time period (CO1)
- Q.8 Write full form of ELCB (CO6)
- Q.9 Define Average value of AC

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- Q.10 Resistance of conductor depends directly on length (T/F) (CO1)
- Q.11 Instrument used to measure electrical energy is (CO1)
- Q.12 Transformation ratio is given by _____ (CO3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 State Fleming's left hand rule (CO3)
- Q.14 Write differences between a.c. and d.c. (CO2)
- Q.15 Draw the circuit diagram of delta connection and write the voltage and current relationship. (CO5)
- Q.16 Draw circuit showing connection of voltmeter, ammeter and wattmeter to measure voltage, current and power across load resistance R_L . (CO1)
- Q.17 Write working principle of transformer. (CO3)
- Q.18 Give the comparison between A.C and D.C motor. (CO1)
- Q.19 What are the precautions to be taken for prevention of electric shock. (CO6)
- Q.20 Explain P type semiconductor. (CO7)
- Q.21 Write applications of electricity. (CO1)

Q.22 Explain colour coding of electrical wires. (CO6)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw and explain the connection of DOL starter for starting 3 phase induction motor (CO5)

Q.24 What is the purpose of earthing? With the help of neat sketch explain the pipe type earthing. (CO6)

Q.25 Draw the constructional details of transformer and label all its parts. Also, write function of each part. (CO3)